

Chapter 3: Conditional Probability and Independence (Sections 3.1 – 3.3)

3.1 Introduction

This section introduces the concept of **conditional probability**, which is crucial for two main reasons:

1. **Partial Information:** It allows us to calculate probabilities when we have some prior knowledge about the outcome of an experiment (e.g., knowing a rolled die is odd affects the probability that it is a 3).
2. **Calculation Tool:** Even without partial information, conditioning often simplifies the computation of complex probabilities by breaking them down into simpler scenarios.

3.2 Conditional Probabilities

Definition

If $P(F) > 0$, the conditional probability of event E given that event F has occurred is defined as:

$$P(E|F) = \frac{P(EF)}{P(F)}$$

- **Intuition:** If we know F has occurred, F becomes our new, reduced sample space. The probability of E occurring is now the probability that the outcome lies in the intersection EF , relative to the size (probability) of the new sample space F .

Computing Probabilities using Reduced Sample Space

When outcomes are equally likely, it is often easier to compute $P(E|F)$ directly by counting outcomes in the reduced sample space F :

$$P(E|F) = \frac{\text{Number of outcomes in } EF}{\text{Number of outcomes in } F}$$

Key Examples:

- **Dice:** If we throw two dice and know the first is a 3 (F), what is the probability the sum is 8 (E)?
 - $F = \{(3, 1), (3, 2), (3, 3), (3, 4), (3, 5), (3, 6)\}$ (6 outcomes).
 - E requires sum 8. Only $(3, 5)$ is in F .
 - Result: $1/6$.
- **Coin Flips:** If a coin is flipped twice ($S = \{hh, ht, th, tt\}$):
 - Prob of two heads (B) given first is head ($F = \{hh, ht\}$): $P(B|F) = 1/2$.
 - Prob of two heads (B) given *at least* one head ($A = \{hh, ht, th\}$): $P(B|A) = 1/3$.
- **Bridge:** If North-South have 8 spades total (leaving 5 spades among 26 cards for East-West), the probability East has exactly 3 spades is calculated using the reduced sample space of 26 cards: $\frac{\binom{5}{3}\binom{21}{10}}{\binom{26}{13}}$.

The Multiplication Rule

Rearranging the definition of conditional probability gives a formula for the intersection of events:

$$P(EF) = P(F)P(E|F)$$

This generalizes to n events:

$$P(E_1 E_2 \cdots E_n) = P(E_1)P(E_2|E_1)P(E_3|E_1 E_2) \cdots P(E_n|E_1 \cdots E_{n-1})$$

Example (Matching Problem Redux):

To find the probability that a specific set of k people have matches (and no one else), we can use the multiplication rule:

$$P(\text{Exactly } k \text{ matches}) \approx \frac{e^{-1}}{k!}$$

3.3 Bayes's Formula

Derivation

Any event E can be decomposed into two mutually exclusive parts: outcomes in F and outcomes not in F (F^c).

$$E = EF \cup EF^c$$

Using Axiom 3 and the Multiplication Rule, we get the **Law of Total Probability**:

$$P(E) = P(E|F)P(F) + P(E|F^c)P(F^c)$$

- **Meaning:** $P(E)$ is a weighted average of the conditional probabilities, weighted by the probability of the conditioning event F .

Bayes's Theorem

This formula is used to "reverse" conditional probabilities—calculating the probability of a hypothesis (F) given evidence (E).

$$P(F|E) = \frac{P(EF)}{P(E)} = \frac{P(E|F)P(F)}{P(E|F)P(F) + P(E|F^c)P(F^c)}$$

General Form (Multiple Hypotheses)

If $F_1, F_2, \dots, F_n$ are mutually exclusive and exhaustive events (partitions):

$$P(F_j|E) = \frac{P(E|F_j)P(F_j)}{\sum_{i=1}^n P(E|F_i)P(F_i)}$$

Important Examples & Applications

1. Medical Testing (False Positives):

- If a test is 95% effective ($P(E|D) = 0.95$) and has a 1% false positive rate ($P(E|D^c) = 0.01$), and the disease is rare ($P(D) = 0.005$):
- The probability of having the disease given a positive test ($P(D|E)$) is only ≈ 0.32 .
- *Lesson:* A high accuracy test can still yield low predictive value if the base rate of the condition is very low.

2. The "Monty Hall" Style Problem (Example 3I):

- Three cards: Red/Red, Black/Black, Red/Black.
- A card is chosen, and one side is Red. What is the probability the other side is Black?
- Many guess $1/2$. The correct answer is $1/3$.
- *Reasoning:* The Red/Red card has 2 red sides, the Red/Black card has 1. It is twice as likely that the visible red side belongs to the Red/Red card.

3. Odds Ratio:

- The odds of an event A are defined as $\frac{P(A)}{P(A^c)}$.
- Bayes's formula in odds form updates beliefs based on the **Likelihood Ratio**:

$$\text{New Odds} = \text{Old Odds} \times \frac{P(E|H)}{P(E|H^c)}$$

- If evidence E is more likely under hypothesis H than H^c , the odds of H increase.